

Physical Chemistry Assignment 2

Instructor

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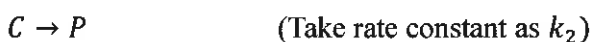
1. The Normalization constant (N) for $\psi = N \sin \frac{n\pi x}{L}$ in the range $0 < x < L$ is $\sqrt{2/L}$

2. In polar coordinates, $\hat{L}^2$ is given by

$$\left[-\hbar^2 \left(\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \sin\theta \frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right) \right]$$

Eigen value of the function $(3\cos^2\theta - 1)$ for the operator $\hat{L}^2$ is $6\hbar^2$

3. Consider the following reaction



$\frac{dP}{dt}$ in terms of reactants and products is $\frac{k_2 k_1 [A][B]}{k_{-1}[D] + k_2}$

4. The de Broglie wavelength of an electron moving with a speed of 2.0×10^6 m/s is

(mass of electron = 9.1×10^{-31} kg) $3.64 \times 10^{-10} \text{ m}$, $3.64 \text{ \AA}$, 0.364 nm

True/False

5. $\cos x$ is an eigen function of operator d/dx . **False**

6. $-i\hbar \frac{\partial}{\partial x}$ is a linear operator. **TRUE**

7. The operator $\hat{p}_x$ commutes with operator $\hat{x}$. **FALSE**

8. In the Haber's process of synthesis of ammonia, temperature is kept higher since the reaction is exothermic. **FALSE**